# Physical Chemistry CUME
# February 5, 2010

The wavefunction of an electron in a ground-state H atom is $\psi = (\pi a^3)^{-1/2} e^{-r/a}$ where $a$=52.9177 pm (picometer=$10^{-12}$ m) and $r$ is the distance of the electron from the proton. Calculate the following probabilities (15 p each):

(a) The electron is in a small cube with sides of 1 pm, located at the nucleus. Note that 1 pm is quite small relative to the size of the H atom, and therefore one can neglect the variation of the electron density within this small cube, and it is OK to assume that the electron density is the same as at the origin (the position of the proton). This way, no complicated integration is necessary.

(b) The electron is in a small cube with sides of 1 pm, at a distance of 1 Å (100 pm) from the proton. See the comments above.

(c) The distance of the electron from the proton is between 49.5 and 50.5 pm. Again, no integral is needed but the problem is different from (a) and (b) because the electron can be anywhere in a 1 pm thick spherical shell with a radius of 50 pm.

(d) The electron is closer to the nucleus than 30 pm.

(e) The electron is further away from the nucleus than 70 pm.

Answer the following question:

(f) What is the distance $r$ where the electron density, i.e. $\psi^2$, is maximum?(10 p)

(g) Someone claims that the correct wavefunction is not $\psi$ above but its negative, $\psi = -(\pi a^3)^{-1/2} e^{-r/a}$. Can he be right? Can you prove that he is wrong? (15 p)

**Help**. You may need the following integrals:

$\int e^{-cx} dx = -e^{-cx}/c$

$\int x e^{-cx} dx = -(cx+1) e^{-cx}/c^2$

$\int x^2 e^{-cx} dx = -(c^2x^2+2cx+2) e^{-cx}/c^3$

Please do not forget that the electron density is the *square* of the wavefunction.

Also recall that the volume element for spherical integration of a spherically symmetrical function is $4\pi r^2 dr$, i.e., the integral of $f(r)$ within a sphere of radius $R$ is $4\pi \int_0^R f(r)\, r^2\, dr$

The surface of a sphere with radius $r$ is $4\pi r^2$.

For question (f), it is helpful to draw a graph showing how $\psi^2$ changes with $r$.